DataCORE

Scientific Storytelling: From Literature Review to Results Language

Warm-up: creating a correlation matrix

A **correlation matrix** shows the pairwise correlation between every pair of numeric variables — usually your **first look** at how the variables in your data move together.

R — `cor()` is built into base R (the `stats` package, loaded automatically — nothing to install):

```r
wages <- read.csv("wages.csv")
cor(wages[, c("experience_years",
              "hourly_wage")])
```

```
                 experience_years hourly_wage
experience_years         1.000000    0.491818
hourly_wage              0.491818    1.000000
```

Python — needs the **pandas** library (`pip install pandas`):

```python
import pandas as pd

wages = pd.read_csv("wages.csv")
wages[["experience_years",
       "hourly_wage"]].corr()
```

🎨 **Hard to read as a table of numbers?** Turn it into a **heatmap** — next slide.

Warm-up: the correlation matrix as a heatmap

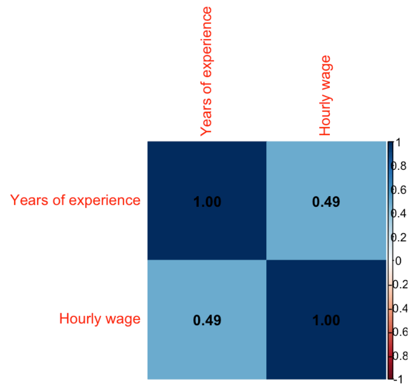
A **heatmap** colors each cell by the strength of the correlation — patterns jump out that a table of numbers hides.

R — needs the **corrplot** library (`install.packages("corrplot")`):

```r
library(corrplot)

vars <- wages[, c("experience_years",
                  "hourly_wage")]
colnames(vars) <- c("Years of experience",
                    "Hourly wage")

corrplot(cor(vars), method = "color",
         addCoef.col = "black")
```



Python — needs **seaborn** + **matplotlib** (`pip install seaborn matplotlib`):

```python
import seaborn as sns
import matplotlib.pyplot as plt

corr = wages.rename(columns={
    "experience_years": "Years of experience",
    "hourly_wage": "Hourly wage"
}).corr(numeric_only=True)

sns.heatmap(corr, annot=True,
            cmap="coolwarm", vmin=-1, vmax=1)
plt.show()
```

`annot=True` / `addCoef.col` print the numbers **on** the colors — best of both worlds.

Afternoon: Scientific Storytelling

1. **The Setup — Your Literature Review as a Story** (1:15–1:45)

2. **Practice — Your Two Slides, Out Loud** (1:45–2:45)

3. **The Payoff — Results Language** (2:45–3:00)

*This morning you learned the **rules** of academic writing. This afternoon you learn the **craft**: turning your literature review into two slides an audience can follow — and your findings into one sentence they can act on.*

1:00 – 1:45 · The Setup

📖 Why storytelling? Because nobody remembers a list

Every good research talk is a **story** with the same beats:

1. **Setting** — here's the world and what we know about it *(background)*
2. **Tension** — but here's what we **don't** know *(the gap)*
3. **Quest** — so we asked this question *(research question)*
4. **Resolution** — and here's what we found *(results)*

💡 Your audience won't remember your fourth citation. They **will** remember the tension: *"we know A and B — but nobody knows C, so we asked…"*

Today we practice the two moments where storytelling matters most: the **setup** (literature → gap → question) and the **payoff** (results).

The funnel: a chapter becomes two slides

Your literature review is **pages** long. Your presentation gets **two slides**:

Slide 1 — Background
Key **definitions + 2–3 cited findings**. What the audience must know to care about your question.

Slide 2 — Gap + Research Question
What's **missing** from what we know — and the **question** that follows from it.

This is not "shrinking" your literature review — it's **distilling** it. You keep only what sets up the question. Everything else stays in the written report.

Anatomy of the Background slide

1. Key definitions (1–2 terms)

- Only terms your audience **needs** — not a glossary.
- One line each, in your own words.

2. What we know (2–3 cited findings)

- Organized by **theme**, not by author.
- One finding per line — **no paragraphs**.
- Each ends with a citation: **(Author, Year)**.

⚠ **The #1 mistake:** pasting sentences from your written review onto the slide. A slide is a **billboard**, not a page. If it takes more than 5 seconds to read a line, cut it.

Worked example — the wildfire PBL case

From this morning's climate case: *Does temperature rise affect wildfire frequency?*

Background: Temperature & Wildfires

Key terms: *Acres burned* = total area destroyed by wildfire per year · *Fire season* = months when large fires occur

What we know:

- Global average temperature has risen about 1.1°C since pre-industrial times (IPCC, 2021)
- Fire seasons in the western U.S. have grown longer and more severe since the 1980s (Westerling, 2016)
- Drought and forest management also shape fire risk — temperature is not the only factor (Abatzoglou & Williams, 2016)

✓ Two terms defined. Three findings, one line each, each cited. A stranger can read this slide in **20 seconds** and be ready for your question.

🧩 Anatomy of the Gap + Research Question slide

1. The gap (1–2 sentences)

- What do the studies you cited **not** answer?
- Point back at your Background slide: the gap should feel **inevitable** after it.

2. The research question (verbatim)

- Word-for-word the question your team wrote — don't paraphrase it live.
- Add your **hypothesis** if you have one.

🔗 The magic sentence connecting the two slides:

"Because we don't yet know ***[gap]****, we ask:* ***[research question]****."*

If that sentence doesn't flow, one of the two slides needs work.

Worked example – Gap + Research Question

The Gap & Our Question

What's missing: Most studies track how **long** fire seasons have become. Fewer connect temperature **directly** to how much land actually burns each year over recent decades.

Our research question: *Is average summer temperature associated with annual acres burned in the United States, 1990–2020?*

Our hypothesis: Years with hotter summers will show more acres burned.

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Say the magic sentence out loud: *"Because we don't yet know whether temperature tracks **acres burned**, we ask: is average summer temperature associated with annual acres burned?"* → It flows – the slides are doing their job.

Four ways these slides go wrong

The wall of text
Full sentences from the written review. Audience reads instead of listens.

The name-drop
"Smith (2020) found… Jones (2021) found…" — organized by author, no theme, no point.

The orphan question
A research question that doesn't follow from the gap — the audience wonders *"why THIS question?"*

The missing citation
A confident claim with no (Author, Year). Instantly costs you credibility.

The fix is the same for all four:

One line per idea. Themes, not authors. Gap points at the question. Every claim cited.

1:45 – 2:45 · Team Task

🏁 Step 1 — Convert your literature review — ~25 min

With your team, convert your literature review into the **two slides**:

1. **Background slide** — 1–2 key definitions + **2–3 cited findings**, one line each, by theme.
2. **Gap + Research Question slide** — the gap in 1–2 sentences, then your RQ **verbatim** (+ hypothesis).
3. Test with the **magic sentence**: *"Because we don't yet know ___, we ask ___."*

Quality bar:

- Could a stranger read each slide in **20 seconds**?
- Is every claim **cited** (Author, Year)?
- Does the question feel **inevitable** after the gap?

25:00

Step 2 — Present to a partner — 10 min

Pair up with someone from **another team**. Each of you:

1. Present your **two slides** in ≤ 2 minutes — *speak to your partner, don't read the slides.*
2. Partner gives **two pieces of feedback**:
   - *"The clearest moment was…"*
   - *"I got lost when…"*
3. Swap roles.

Listeners: you are the audience test. If you can't repeat back the gap and the question after 2 minutes, say so — that's the most useful feedback your partner will get today.

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10:00

Back with your team: apply the feedback to the slides **now**, while it's fresh. These two slides go straight into your final presentation.

2:45 – 3:00 · Scientific Storytelling

Two registers — you need **both**

The number (for credibility)
"The regression coefficient on experience was 0.34 ($p < 0.05$)."

Other researchers can **check** this.

The plain language (for the audience)
"Our analysis found that workers earned about 34 cents more per hour for each additional year of experience."

A policymaker can **act** on this.

⚖️ Strong results writing puts the **plain sentence first**, then backs it with the numbers in parentheses. Never make the reader decode statistics to learn what you found.

ℹ️ You'll learn to **produce** numbers like these in Sessions 8–9. Today you learn how to **say** them.

Translation practice — say it both ways

Each statistical sentence below is **true but unusable** on its own. Translate it out loud:

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→ *"Our analysis found that **hotter summers went hand in hand with more land burned** — a fairly strong association (r = 0.62). Association, not proof of cause."*

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→ *"Our analysis found that **hotter summers went hand in hand with more land burned** — a fairly strong association (r = 0.62). Association, not proof of cause."*

Notice the pattern: **plain claim first** (direction + size in real units), **statistics in parentheses**, **caveat if needed**.

Draft your team's headline finding — now

Write **one results sentence** for your team's main finding, in this shape:

> *"Our analysis found that **[plain-English finding]** (**[stat]** = __)."*

Then read it to a partner who is **not** on your team. Did they get it?

*No statistics yet? Use your strongest **descriptive** finding — a difference, a trend, a pattern — and put its number in the parentheses.*

Checklist:

- Plain-English claim **first**?
- Direction clear (more / less / higher)?
- The number **with units**?
- Statistics in parentheses?
- No causal verb unless you can defend it?

10:00